



Ace's Realm

X - P L A N E 1 2

Modlist & Flight Guide

Target version 12.4.3 · Six waves · 2026

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Installation Guide

Game Prep

This guide targets **X-Plane 12** version **12.4.3** (the “VR Update”, released 2026-06-25). Install X-Plane 12 from Steam or directly from the Laminar Research store, let it finish its first-time setup, then launch it once to the main menu so the sim generates its folders and preference files.

1. Install **X-Plane 12** from Steam or the Laminar store and let it download completely.
2. Launch the sim once, all the way to the main menu, then quit.
3. Confirm the version line reads **12.4.3** (shown on the main menu / About screen). This is the version this entire guide — and every mod you add to it — is written against.

Note: Version numbers change whenever the sim updates. Every time X-Plane 12 downloads an update, re-check the version line. If it no longer reads 12.4.3, treat every mod in this guide as unverified until you re-check its page date (see **Version Compatibility** below). As of writing, 12.4.4 and 12.5.0 were still planned releases — plan around them the same way.

Tip: X-Plane 12 is a large download (tens of GB). Give it room on an SSD if you have one; scenery loading and ortho (see AutoOrtho in the Foundational chapter) benefit most from fast disks.

Where Things Live in X-Plane 12

X-Plane 12 keeps its world in one root folder. Learning this map makes every install step in this guide understandable:

1. Aircraft/ — the aircraft library. Each aircraft is a subfolder with its .acf file, cockpit objects, sounds, and docs.
2. Resources/plugins/ — where plugin folders go. Most of the foundational tools in this guide (AviTab, X-Camera, X-ATC-Chatter, openSAM, Better Pushback, and others) drop in here.
3. Custom Scenery/ — where scenery packs and libraries live, plus the scenery_packs.ini file that decides their load order.
4. Custom Data/ — navigation data (navdata/AIRAC cycles) that overrides the sim’s default data. Navigraph writes here.
5. Output/ — logs, preferences, and saved flight plans. Log.txt (sim root) is the log the manager can analyze.
6. **Steam vs direct:** a Steam install lives at ...\\Steam\\steamapps\\common\\X-Plane 12\\; a direct Laminar install lives wherever you chose at install time (often D:\\X-Plane 12\\). Both work identically — point the mod manager at whichever one you have.

Warning: Never install a plugin by unzipping the **contents** of its folder straight into Resources\\plugins — each plugin needs its own subfolder there (e.g., Resources\\plugins\\AviTab\\). Dropping loose files into the plugins folder breaks the sim’s plugin loader. The mod manager handles this for you; when installing by hand, copy the folder, not the files.

Where Mods Come From

The two hubs used for this modlist’s ecosystem, as supplied:

1. **X-Plane.org** — the long-standing community forums and store, with both free files and payware — <https://forums.x-plane.org/> (store at <https://store.x-plane.org/>).

2. **X-Plane.to** — a modern hub focused on free mods, liveries, and add-ons (similar to Flightsim.to) — <https://x-plane.to/>.

Steam Workshop is not a major source for X-Plane 12. Whenever a mod card in this guide carries a URL, that URL is your source of truth; when it does not (yet), the card says so.

XFast Manager — the Mod Organizer

The organizer for this modlist is **XFast Manager**, an open-source (GPL-3.0) X-Plane addon installer and manager built with Tauri (Rust + Vue) by developer CCA3370 — github.com/CCA3370/XFast-Manager. It ships for Windows 10/11 (x64), macOS 10.15+ (Intel and Apple Silicon, universal builds), and Linux (x64; GTK3 + WebKit2GTK required).

Its advertised feature set, straight from the project README:

1. **Install in seconds** — drag and drop any addon file, archive, or folder.
2. **Zero guesswork** — the app identifies addon types and puts them in the right place.
3. **Stay organized** — manage aircraft, plugins, navdata, scenery, and liveries in one view.
4. **Safe and reversible** — backups, verification, and clear conflict warnings.
5. **Smart scenery sorting** — keeps the scenery load order clean and stable.
6. **Update awareness** — see update availability at a glance.
7. **X-Plane log analysis** — a built-in view for reading the sim's Log.txt.
8. **Polished UI** — light/dark themes, English/Chinese support, and (on Windows) a right-click install menu you can enable in Settings.
9. **Archive support** — installs directly from ZIP, 7z, or RAR, or from folders, without manual extraction.

Note: Facts about XFast Manager in this chapter (features, versions, asset names) come from the project's GitHub README and release notes, verified 2026-08-05. The latest release at the time of writing is **v1.2.5** (2026-05-31). If a newer release exists, prefer it — the steps below do not change.

Installing XFast Manager

Open the [releases page](#) and download one file for your system:

1. **Windows** — recommended: XFast-Manager-<version>-windows-setup.msi (an installer). Prefer the .msi for a normal install; if your browser blocks .msi, use the windows-setup.zip (same installer inside). A windows-portable.zip is also offered if you want a no-install copy.
2. **macOS** — XFast-Manager-<version>-macos-universal.zip (works on both Apple Silicon and Intel Macs).
3. **Linux** — XFast-Manager-<version>-linux-x64.AppImage for most distributions, or the .deb / .rpm / Arch tar.gz packages for those systems.

Run the installer (Windows/macOS) or launch the AppImage/binary (Linux — install the gtk3 and webkit2gtk-4.1 packages if it does not start). The app does not need to live next to X-Plane 12; it only needs to know where the sim is.

First Launch & Selecting the X-Plane Folder

On first launch, XFast Manager asks you to select your X-Plane folder. The app manages everything through that one root folder:

1. Confirm the folder it is pointed at. For Steam that is ...\\Steam\\steamapps\\common\\X-Plane 12\\; for a direct Laminar install, the folder you chose at install time.
2. Confirm the main views make sense: aircraft, plugins, navdata, scenery, and liveries should all populate as the app scans the folder. (Scanning may take a moment on large installs.)

3. Set the theme you like (light/dark) and enable the Windows right-click install menu in Settings if you want it.

Note: The README describes selecting the X-Plane folder manually on first launch. If your installed version also offers automatic detection, use whichever works — the requirement is only that the app points at the **root** of the X-Plane 12 folder, not a subfolder inside it.

Installing Mods

Every mod in this guide is listed as a **mod card** in its chapter (the Foundational chapter for the base plugins, per-wave chapters for wave content), and every card carries the fields this project's rules demand. To bring a mod into XFast Manager:

1. **Drag & drop** — download the mod as .zip, .7z, or .rar (or leave it as a folder) and drop it onto the XFast Manager window. The app identifies the addon type and puts it in the right place — aircraft to Aircraft/, plugins to Resources/plugins/, scenery to Custom Scenery/, navdata to Custom Data/, liveries to the matching aircraft's liveries/ folder.
2. **Protected archives** — if the archive asks for a password, provide it when prompted.
3. **Conflict resolution** — when a file already exists, the app offers a clean install or an overwrite; enable verification for safety. Choose deliberately, and record any overlap you care about in conflicts-mods.md (that file is deliberately kept out of the PDF so you can maintain it freely).

Warning: Close X-Plane 12 during installs. The sim holds file locks on its folders while running, and the README's own troubleshooting section lists permission failures from exactly this. Quit the sim, install, then relaunch.

Managing Mods

1. **One view for everything** — aircraft, plugins, navdata, scenery, and liveries are managed from the same interface with quick actions (enable/disable-style housekeeping, open folder, remove).
2. **Update availability** — the manager shows at a glance which installed items have newer versions available; check it after every sim update.
3. **Log analysis** — after a session, use the log-analysis view on Log.txt to see what loaded (or failed to load). This is the fastest way to confirm a plugin actually started.

Scenery Load Order

X-Plane 12 reads Custom Scenery/scenery_packs.ini top-to-bottom: entries higher in the file draw on top of entries lower down. The general order that works for this guide:

1. **Airports** (payware and freeware airports first).
2. **Overlays** (things that must sit on top of terrain: roads, forests, objects).
3. **Ortho / mesh** (AutoOrtho output and ortho tiles — they must sit **below** airports and overlays or they bury them).
4. **Libraries** (OpenSceneryX-style shared object libraries, when used) and the sim's global scenery at the very bottom.

XFast Manager's **smart sorting** maintains this order for you. Treat it as the default; if you ever hand-edit the .ini (or reinstall the sim and lose it), rebuild it with the manager and re-verify the order above.

Warning: An airport hidden **below** an ortho tile in the .ini will not appear — it is one of the most common “my airport vanished” causes. If an airport or library disappears, check the .ini order before anything else.

Backup & Restore

1. **Mods** — XFast Manager's backup support covers the managed addons themselves (backups plus verification when installing). Keep the mod cards in this guide as the human-readable record: each card's fields are enough to re-install everything.
2. **Saves & preferences** — X-Plane 12 keeps preferences and controller profiles under `Output/preferences/` in the sim root; copy that folder aside before switching waves or updating several mods at once.

Version Compatibility

This guide is written for version **12.4.3**, and every mod in it must be compatible with that version. The heuristic for judging compatibility, when a specific mod version is unknown, is the **patch / release date**:

1. A mod whose page shows it was updated close to the release of 12.4.3 (2026-06-25) is the safest bet — it was most recently checked against the current sim.
2. A mod that has not been touched in months may still work fine, but it is riskier.
3. Dependencies matter as much as dates: a mod that requires something you do not have (a library, a subscription, a specific other plugin) is incompatible no matter how fresh it is.

Tip: Read each mod's own page before adding it — the page's last-updated date and listed requirements are the signals to trust. And whenever the sim updates, re-run the manager's update check and re-read the affected mod pages: a new sim build can break plugins that were fine the day before.

Launch & Verify

1. Start X-Plane 12.
2. Check the version line reads 12.4.3.
3. Confirm the expected plugins loaded (the sim's Plugins menu lists loaded plugins; the manager's log analysis confirms it too).
4. Fly the wave's opening flight and confirm the mods behave the way the guide describes — in-sim behavior is the ground truth.

Tip: Do not trust a mod list from memory. XFast Manager's views are the authoritative picture of what is installed — one glance before each session tells you the wave is set up right.

Per-Wave Switching

Each wave is its own new-game setup with its own aircraft and mod subset. Switching waves is a manager chore, not a reinstall:

1. Open XFast Manager and bring up the wave's loadout (the wave chapter's mod cards are the checklist).
2. Enable that wave's content and keep the Foundational plugins (the base tools from the Foundational chapter) available — they are installed once and used across all waves.
3. Start the wave's flight with its aircraft selected.

Warning: The wrong mods break the wave design. If content intended for a later wave is carried into an earlier one, the intended progression is lost and the sim behaves differently than the guide describes. Review what is active before each new wave.

Tip: Because the Foundational plugins never move, per-wave switching only ever touches wave content. If the manager's version supports named save states or profiles for the mod set, use them; otherwise the wave cards are the checklist.

Foundational Plugins & Tools

The Base Toolset

This chapter documents the **foundational plugins and tools** of Ace's Realm. They are the base of the modlist: installed once, available in every wave, and assumed by the wave guides. They were supplied as a group — Navigraph, LittleNavMap, AutoOrtho, X-Camera, AviTab, X-ATC-Chatter, openSAM, and Better Pushback — and each entry below explains what it does, how to install it, and how to use it.

Note: Provenance. The usage and installation prose in this chapter is general guidance written from general knowledge, not verified against each tool's current release — per this project's rules, mods are not researched here. Exact versions, current menu labels, and compatibility with your 12.4.3 install are yours to confirm from each mod's own page (use the patch/release-date heuristic from the Installation chapter). The download links below were added at your request and verified to resolve on 2026-08-05 — each points to the tool's official site or repository, which is also where you confirm the current version.

Navigraph (subscription & setup)

Navigraph is a subscription service that keeps your simulator's navigation data and charts current. It is less a single mod and more a **data and planning service** that several other tools (AviTab below, and any FMS-equipped aircraft) plug into.

What it does — two main products:

1. **Navigation data (AIRAC cycles)** — real-world navigation databases (waypoints, airways, airports, procedures) refreshed on a 28-day cycle. Without a subscription, a sim's navdata ages and drifts from the real world; with it, your FMS and GPS match today's charts.
2. **Charts** — professional Jeppesen-style instrument charts (SIDs, STARs, approaches, airport diagrams) delivered via web, desktop/mobile apps, and in-sim through AviTab.

Install & setup:

1. Create an account at the service's site and take a subscription (the tiers differ in products and device limits — pick the one that covers your sims).
2. Install the **Navigraph Hub** application (the account/launcher/update center) from the official site — navigraph.com.
3. Inside the Hub, install the **FMS Data Manager** component, tell it about X-Plane 12, and download the current AIRAC cycle. It writes the data into the sim's Custom Data/ folder.
4. For charts in the cockpit, see AviTab below — with your account linked, AviTab's tablet can show your Navigraph charts.
5. Keep the subscription active and refresh the cycle each month (the Hub notifies you); every 28 days a new cycle lands.

Wave fit — Wave 0 can learn with the sim's default data; the subscription becomes valuable from Wave 1 onward (flight planning) and is essentially assumed by Waves 2-5 (IFR, FMS, airline ops).

Navigraph

Dependencies: user to provide

Impact: Replaces/updates the sim's navigation data with current AIRAC cycles and provides real instrument charts (also in the cockpit via AviTab).

Notes: Subscription-based; monthly data refreshes; writes to the sim's Custom Data folder. Current version and any dependencies: user to provide.

LittleNavMap

LittleNavMap is a free, external (separate-window) flight planning and moving-map application for Windows, macOS, and Linux. It runs **outside** X-Plane 12 and connects to it over the network for live aircraft position.

What it does — a full map of the world with airports, nav aids, airspace, and procedures; flight planning tools that export plans the sim can load; and a live moving map of your aircraft while you fly. It is the planning companion the wave guides reference.

Install:

1. Download the latest release from the official site — littlenavmap.org — and extract the archive to any folder — no installer needed.
2. Launch the application (e.g., `LittleNavmap.exe` on Windows).

Setup & usage:

1. **Simulator database** — on first start, point it at your X-Plane 12 folder so it reads the sim's scenery/nav database (it needs to know where the sim is to show airports and procedures correctly).
2. **Live connection** — enable live position reporting: in X-Plane 12's Data Output settings, turn on the UDP data items (position, attitude, etc.), and in LittleNavMap's simulator connection settings match the IP/port (local loopback and the sim's UDP data port are typical defaults; confirm the exact labels in the current documentation).
3. **Plan a flight** — pick a departure and destination, let it build a route, review it, and export it in the sim's flight-plan format (`.fms`). Load the plan in X-Plane 12 (the sim's flight-plan load feature reads plans saved to its output folder).
4. **Fly** — with the connection live, your aircraft shows as a moving symbol on the map, with track, altitude, and progress against the plan.

Wave fit — a Wave 0 learning aid (see where you are), essential from Wave 1 onward for cross-country and IFR planning, and used by every later wave.

LittleNavMap

Dependencies: user to provide

Impact: External flight planning and moving map; exports flight plans the sim loads; live-tracks your aircraft over UDP.

Notes: Freeware; runs outside the sim; needs the sim's UDP data output enabled for live tracking. Current version and any dependencies: user to provide.

AutoOrtho

AutoOrtho streams orthophoto imagery (real satellite/aerial photos) into X-Plane 12 **on demand**, instead of downloading terabytes of ortho scenery in advance. Where it has tiles, the ground looks photoreal; elsewhere the sim's default scenery still shows.

What it does — a launcher app and a scenery link. You pick imagery sources and regions; it downloads and caches tiles as you fly over them, and presents itself to the sim as a normal scenery pack.

Install & setup:

1. Download the AutoOrtho application from its official repository — github.com/kubilus1/autoortho (the current build is on the releases page) — and install it.

2. Point it at your X-Plane 12 folder and choose an imagery source and region(s) you fly in; let it build/download the tile set (the first download is the big one — after that it streams).
3. The app installs/links its scenery entry into Custom Scenery/ — make sure it sits in the ortho/mesh position of the load order: **below** airports and overlays (see Scenery Load Order in the Installation chapter). XFast Manager's smart sorting handles this.
4. Start the AutoOrtho launcher **before** X-Plane 12 each session, then fly normally.

Tip: Cache and bandwidth: AutoOrtho needs a decent internet connection and a few dozen GB of disk cache for the regions you fly. Pre-download regions you fly often; let the cache grow for everywhere else. If the ground flickers or falls back to default textures, the scenery order or the cache is the first thing to check.

Wave fit — optional but transformative from Wave 1 on, where cross-country flights over real terrain shine. Not needed to learn in Wave 0.

AutoOrtho

Dependencies: user to provide

Impact: Streams photoreal ortho ground textures into the sim on demand (scenery/visual only; no flight-model or system changes).

Notes: Requires internet while flying, disk cache, and correct scenery load order (below airports/overlays). Launcher must run before the sim. Current version and any dependencies: user to provide.

X-Camera

X-Camera is a freeware camera system plugin. The sim's default views work, but X-Camera gives you **your own** saved camera positions with smooth transitions — cockpit close-ups, wing views, tower-style chase shots, and more.

What it does — define named cameras anywhere around the aircraft, assign them to joystick buttons/keys, move between them with animated transitions, and save per-aircraft camera sets.

Install:

1. Download X-Camera from the developer's site — stickandrudderstudios.com/x-camera (a free version is available there) — and extract the plugin folder.
2. Copy the plugin folder into X-Plane 12/Resources/plugins/ — either by hand (folder, not contents) or by dropping the archive onto XFast Manager.

Setup & usage:

1. **Bind controls** — in X-Plane 12's Settings → Keyboard/Joystick, search for the X-Camera commands and bind what you want: enable camera control, next/previous camera, and transition on/off. This is the one-time chore that makes everything else quick.
2. **Build cameras** — with camera control active, fly the aircraft to a view you like (pan with the mouse/controls), then save it as a camera in the X-Camera configuration panel. Do this for a pilot's view, a co-pilot view, a wing view, a tail chase, etc.
3. **Assign & save** — assign each camera to a button/key, then save the camera set for that aircraft (X-Camera keeps per-aircraft sets, so each plane remembers its own cameras).
4. **Fly** — tap through your cameras mid-flight with the bound buttons.

Wave fit — pure quality of life; useful from Wave 0 (find the best instructional views) through Wave 5 (external views of the big freighter).

X-Camera

Dependencies: user to provide

Impact: Custom camera system: saved views, transitions, per-aircraft sets; camera/immersion only.

Notes: Bindings are set in the sim's Keyboard/Joystick settings; per-aircraft camera sets save with each aircraft. Current version and any dependencies: user to provide.

AviTab

AviTab is a freeware plugin that puts a tablet in the cockpit. It shows PDFs (approach plates, manuals, checklists) and maps, and can display Navigraph charts when your account is linked.

What it does — an in-cockpit moving map and PDF viewer: put your charts and manuals in AviTab's folder, open the tablet in the sim, and page through them without leaving the cockpit.

Install:

1. Download AviTab from its repository — github.com/TeamAvitab/avitab (the active home; downloads are on the releases page) — and extract the plugin folder.
2. Copy it into X-Plane 12/Resources/plugins/ (by hand or via XFast Manager).

Setup & usage:

1. **Add documents** — copy PDF charts/manuals into AviTab's document folder inside the plugin directory (Resources/plugins/AviTab/...). Organize them in subfolders per region or aircraft.
2. **Open the tablet** — bind a key/joystick button to the AviTab command in the sim's Keyboard/Joystick settings; aircraft with a built-in 3D tablet open it by touching the tablet; the key command is the universal fallback.
3. **Charts** — if you have a Navigraph subscription, link the account and pull current charts straight into the tablet (cross-reference the Navigraph section above).
4. **Maps** — the tablet can show a moving map of your position with the charts loaded.

Wave fit — the classic Wave 0 companion (checklists and airport diagrams on a tablet) and indispensable from Wave 1 on for approach plates.

AviTab

Dependencies: user to provide

Impact: In-cockpit tablet: PDF charts/manuals, moving map, optional Navigraph charts integration.

Notes: PDFs go into the plugin's document folder; bind a key/button to open it in aircraft without a 3D tablet. Current version and any dependencies: user to provide.

X-ATC-Chatter

X-ATC-Chatter is a freeware ambient-radio plugin. It plays realistic, location-aware background ATC/company radio chatter so the frequency feels alive — even where there is no real controller.

What it does — generates simulated radio traffic (clearances, position reports, other aircraft talking to “their” controllers) based on your region and phase of flight, as ambient audio. It does not replace or control the sim's ATC; it layers believable background noise on top.

Install:

1. Download X-ATC-Chatter from the developer's site — stickandrudderstudios.com/x-atc-chatter-1-7-2 — and extract the plugin folder.
2. Copy it into X-Plane 12/Resources/plugins/ (by hand or via XFast Manager).

Setup & usage:

1. Open the plugin's configuration panel from the sim's Plugins menu.

2. Choose the chatter content/volume you like — the plugin ships with default chatter; tune the volume so it sits under your own radio calls.
 3. Fly. Chatter follows your situation: busier near airports, quieter en route. Toggle it when you want silence (e.g., while practicing instrument procedures where you must hear the sim's ATC clearly).
- Wave fit** — immersion from Wave 0 onward; most at home in Waves 1-3, where radio work is constant.

X-ATC-Chatter

Dependencies: user to provide

Impact: Ambient background ATC/company radio chatter (immersion/audio only; does not change the sim's ATC or flight model).

Notes: Configure volume and packs in the plugin menu; mute it when you need to hear the sim's real ATC. Current version and any dependencies: user to provide.

openSAM

openSAM is the open-source scenery-animation system you supplied — github.com/hotbso/open-SAM. It provides the **Scenery Animation Manager**-style features for airports that support them: animated jetways, docking guidance, marshallers, and ground service animations — as a free, open-source project.

What it does — at compatible airports, animated jetways extend to your aircraft, marshallers guide you to the stand, and ground services animate around you. It fills the “the airport is alive” gap that static scenery leaves.

Install (follow the project's own instructions from the supplied URL; general outline):

1. Download the latest release from the project's releases page (linked above).
2. Install the plugin part into X-Plane 12/Resources/plugins/ and the library part into X-Plane 12/Custom Scenery/ as the project's install notes direct.
3. Confirm the scenery library sits in the **libraries** position of the load order (see Scenery Load Order in the Installation chapter) — XFast Manager's sorting handles it.

Usage — mostly automatic: at airports built with SAM-style animation, jetways/marshallers respond to you pulling up to the stand. Pair it with Better Pushback below for a complete ground experience.

Wave fit — from Wave 0's home-field circuits to Wave 5's cargo terminals; it shines wherever you park at a gate.

openSAM

Dependencies: user to provide

Impact: Open-source SAM-style scenery animation: animated jetways, docking guidance, marshallers, ground services at supporting airports.

Notes: URL supplied by the user. Plugin + library install (see the project's own instructions); library belongs in the scenery load order. Current version and any dependencies: user to provide.

Better Pushback

Better Pushback is a freeware plugin that gives you a proper pushback tug with a route you draw, instead of the sim's default straight-line push.

What it does — press the pushback command, and a tug appears and connects; drag a path on the screen and the tug pushes the aircraft along it, steering exactly where you drew, then disconnects when done. You stay in control of the ground flow at gates and ramps.

Install:

1. Download Better Pushback from its repository — github.com/skiselkov/BetterPushbackC (downloads are on the releases page) — and extract the plugin folder.
2. Copy it into X-Plane 12/Resources/plugins/ (by hand or via XFast Manager).

Setup & usage:

1. **Start** — trigger the pushback command (a default key is assigned by the plugin; rebind it in Settings → Keyboard if you prefer).
2. **Draw the route** — with the tug connected, click/drag on the screen to lay out the path (clear of obstacles, out of the stand).
3. **Watch it work** — the tug follows the route, stops, and disconnects; you release the parking brake, start engines, and taxi normally.
4. **Preferences** — the plugin keeps its settings in the sim's preferences; adjust behavior there (e.g., auto-start, volume of the tug's sounds) as you like.

Wave fit — a Wave 0 convenience (parking and pushing at the school field) that becomes a daily ritual in Waves 3-5 at big airports.

Better Pushback

Dependencies: user to provide

Impact: Interactive pushback tug with a drawn route (ground handling/QoL; no flight-model changes).

Notes: Trigger command and rebind in the sim's keyboard settings; pair with openSAM for a full gate experience. Current version and any dependencies: user to provide.

Where They Fit

The eight tools divide naturally into **planning** (Navigraph, LittleNavMap), **cockpit** (AviTab, X-Camera), **immersion** (X-ATC-Chatter, openSAM, Better Pushback), and **visuals** (AutoOrtho). Every wave assumes the base toolset is installed; the wave chapters tell you when a specific tool becomes central — LittleNavMap and Navigraph from Wave 1, AutoOrtho's photoreal terrain from Wave 1's cross-country legs, and the ground/animation trio every time you fly a gate.

Waves & New-Game Setups

Wave 0: First Solo

How to Play

This chapter is the complete beginner tutorial for Wave 0. It assumes you have never flown in X-Plane 12 and are flying the **vanilla sim** — the only additions are the foundational tools from the Foundational Plugins chapter, which make the sim easier to see, operate, and plan. Wave 0 exists to teach you to fly; the mods you add to it later only make that clearer.

The Story So Far

First Solo. Cedar Harbor Flight School, a one-runway field on a grey-blue coast, has a new student: you. Your instructor has watched you fly the sim's built-in lessons, and today she hands you the keys to the school's Cessna 172 — white, patient, and absolutely not impressed by you yet. Your mission, should you choose to accept it (you already have): one complete traffic pattern, takeoff to landing, without the instructor in the right seat. There is a thermos of coffee in it for you at the end. The pattern is a mile of sky, three turns, and one very personal moment on short final.

Before You Fly

1. **Controls** — a joystick (or yoke + rudder pedals) makes a different game of it; a mouse and keyboard work, but the school will notice the difference. Calibrate in Settings → Joystick, and check the response feels linear.
2. **Graphics & realism** — run Settings → Graphics to a frame rate you can read gauges at. Leave the flight-model realism settings at the sim's defaults for Wave 0; “easy” assists hide the lessons you came for.
3. **Where to start** — use the sim's Flight Configuration to place yourself on the runway at your home field (start “on the runway” for your first attempts — cold & dark can wait for Wave 1).
4. **Views** — bind the view controls you like (Settings → Keyboard): the sim's defaults cover panning and cockpit/external cycling. AviTab's tablet (Foundational chapter) is your charts and checklist holder; X-Camera adds saved custom views when you want them.

The Cockpit in One Page

The 172's panel tells you everything you need with a handful of instruments:

1. **Airspeed** — the tape on the left (PFD) or round gauge: how fast you are moving through the air.
2. **Attitude** — the artificial horizon: the single most important instrument. Wings level, horizon on the horizon.
3. **Altitude** — the barometric altimeter: how high above sea level you are (set QNH from the local weather).
4. **Heading** — the compass/heading indicator: which way you point.
5. **Vertical speed** — the VSI: whether you are climbing or descending, and how fast.
6. **Throttle (blue) / Mixture (red) / Flaps (lever)** — power, fuel-air, and high-lift devices. In the 172: throttle forward = more power, mixture rich = correct fuel, flaps extend for slow approaches. Trim is your friend from flight one: hold the nose where you want it, then take the pressure off with the trim wheel so you can fly with fingertips, not fists.

Your First Flight

A complete circuit, step by step. Numbers are typical 172 values — your aircraft's documentation (and AviTab with the POH PDF) is the authority:

1. **Power on** — battery/master on, prime if needed, mixture rich, throttle a touch, starter until the engine catches, then idle check.
2. **Taxi** — release the parking brake, throttle up slightly, and steer with the rudder pedals (not the yoke). Brakes slow you; keep the speed walking-pace.
3. **Run-up** — at the run-up area: brakes set, throttle to about 1700-1800 rpm, check magnetos and gauges per the checklist, then idle back.
4. **Line up** — call the traffic pattern position (see Radio Basics), line up on the centerline, apply full power smoothly.
5. **Rotate** — around 55-60 knots, pull back gently; the 172 lifts off and wants to climb — hold about 70-75 knots on climb with the nose pitched for the attitude.
6. **Crosswind turn** — at about 500 ft above field elevation, turn downwind; level off at pattern altitude (typically 1,000 ft AGL, the field's published pattern altitude), power to cruise.
7. **Downwind checks** — once abeam the threshold: carb heat on, power back, flaps 10-20° as the checklist says, pitch for approach speed.
8. **Base** — turn base when the threshold is 45° behind your wing; descend, flaps to full (30-40° in the 172), aim for 65-70 knots on final.
9. **Final** — line up with the runway, keep the aim point fixed (not moving in the windshield), power for the speed, and hold the glide.
10. **Flare & land** — over the threshold, round out: pull the nose up just enough that the sink stops, let the mains touch, hold the nose off as long as you can, then let it down gently.
11. **Rollout & clear** — brakes to taxi speed, exit at the first taxiway you can, and stop to collect your thermos.

Tip: If a landing goes wrong, do not try to force it. Go around: full power, nose up, flaps up a notch at a time as speed builds, climb back to pattern altitude, and set up again. Going around is not failure — it is the skill. Wave O's real goal is a pattern where going around is a calm choice, not a rescue.

The Traffic Pattern

The pattern is the grammar of all aviation. Learn it by feel in Wave O and every later wave inherits it:

1. **Upwind** — after takeoff, climbing straight out from the runway.
2. **Crosswind** — the 90° turn off the runway heading.
3. **Downwind** — parallel to the runway, opposite direction, at pattern altitude (usually 1,000 ft AGL).
4. **Base** — the descending turn across the runway's threshold.
5. **Final** — the straight-in approach to landing, at approach speed with full flaps.

Fly the same numbers every time — same altitude, same speeds, same turn points. The pattern should feel like a dance, not a negotiation.

Radio Basics

1. **Non-towered field** — announce on the CTAF/UNICOM frequency: who you are, where you are, what you are doing ("Cedar traffic, Cessna one-seven-two, downwind, runway two-four, Cedar").
2. **Towered field** — call ground for taxi, tower for takeoff/landing; read back clearances. X-Plane 12's built-in ATC does this with you — turn it on and follow it.
3. **ATIS** — listen first at controlled fields: it gives you the wind, the runway in use, and the altimeter (QNH) setting.

Note: With X-ATC-Chatter installed (Foundational chapter), the frequency will sound alive around your own calls — that is ambient background audio, not the sim’s ATC. Keep it quiet enough to hear the real controller; mute it when practicing instrument work later.

Staying Alive: Speed, Fuel, Altitude

1. **Speed** — slow is not always safe: too slow is a stall (the wing stops flying). Keep the approach speeds above; if the stall warning sounds, lower the nose, add power.
2. **Fuel** — the 172 carries hours of fuel; still, glance at the gauges each lap. A full tank beats a creative forced landing.
3. **Altitude** — in the pattern, altitude is a commitment: below pattern altitude on downwind is a problem; on final, the aim point tells you if you are high or low — if it slides down the windshield, you are sinking; if it slides up, you are floating.

Using the Sim’s Tools

1. **Flight Configuration** — start on a runway, in the air, or cold & dark; pick the aircraft and the airport. Use “start on runway” to practice circuits without the full start flow.
2. **Time warp** — speed up long cruise legs so the flying stays interesting; pause anytime (the sim pauses on its own when you lose focus, by default).
3. **ATC window** — the sim’s ATC menu drives clearances and traffic; use it from Wave 0 so the phraseology starts to stick.
4. **Camera & tablet** — cycle views (default view keys) for situational awareness; AviTab holds the airport diagram and checklist; X-Camera’s saved views are your personal best angles.
5. **Instant replay & screenshots** — the sim can replay your last seconds; use it after a rough landing to see what your hands actually did.

New-Game Checklist

1. Confirm the main menu reads version 12.4.3.
2. Confirm XFast Manager shows the foundational toolset installed (and nothing else yet).
3. Start a new flight in the Cessna 172, on the runway at your home field, clear skies.
4. Fly one complete pattern: takeoff, downwind, base, final, landing.
5. Repeat until one pattern feels calm — that is your solo.
6. When you can hold altitude and speed without fighting the controls, Wave 1’s cross-country legs are waiting.

Wave 0: Modlist

Wave 0 adds no gameplay content. Everything in this wave exists to make the sim easier to **see**, easier to **operate**, or easier to **plan**. Nothing here changes an aircraft’s performance, the weather, or the flight model — that is the wave’s hard rule. Wave 0 admits only **UI and quality-of-life** mods, and the foundational toolset from the Foundational Plugins chapter is the entirety of it for now.

The active Wave 0 loadout is therefore:

1. The **foundational toolset** (Foundational Plugins chapter): Navigraph (optional), LittleNavMap, AutoOrtho (optional), X-Camera, AviTab, X-ATC-Chatter, openSAM, Better Pushback. These are installed once and stay available in every wave.

Beyond that, the cards below are **pending addition**: per project decision, this project does not research, select, or validate mods — you do. When you are ready, replace a placeholder card with one mod card per mod. Each card needs the fields this guide’s rules demand: a clickable name with URL, dependencies, system/mechanic impact, and notes. Anything you do not supply stays marked “**user to provide**”.

The mods you add split across the next two sections by what they change: **Mechanics** for tools that change how you work with the sim, and **Graphics** for mods that change how it looks.

Mechanics (UI & Quality-of-Life)

Tools that make the sim easier to operate and plan — navigation aids, information displays, ground-handling conveniences — without changing any aircraft's numbers.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Graphics (Visual & Interface)

Mods that only change how the sim looks or reads: textures, colors, HUD styling, readability. Nothing here changes behavior.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Wave 0: Mechanics

The systems you meet in Wave 0 are the ones every later wave builds on. Learn them in the 172, where they are simple and unhurried.

The Primary Instruments

1. **Airspeed indicator** — speed through the air. In the pattern, this is your primary management tool: too slow and the wing stops flying; too fast and you float forever.
2. **Attitude indicator** — the artificial horizon. Pitch and bank are read off it; fly the horizon, trust it over your stomach.
3. **Altimeter** — height above sea level, set to QNH. In the pattern, altitudes are the contract with everyone else in the circuit.
4. **Heading indicator** — compass heading. Turn it, then correct for drift on windy days (crabbing into the wind).
5. **Vertical speed indicator** — climb/descent rate. Lags a moment; use it to confirm the attitude you set, not to chase it.
6. **Turn coordinator** — rate of turn; the standard-rate turn (2 minutes per 360°) is the polite turn in instrument flying later.

Powerplant Basics

1. **Throttle** — power. Pull it back and the airplane slows and wants to descend; push it up and it accelerates and climbs.
2. **Mixture** — fuel-air ratio. Rich for takeoff and climb, leaned for cruise. Too lean at full power is a hot engine; too rich at altitude is rough running.
3. **Flaps** — lift and drag on demand. Extending flaps lowers the stall speed and steepens the descent for the approach; retracting them cleans the airplane up for climb.
4. **Fuel management** — tank selector, gauges, and the simple rule: know which tank feeds the engine and how much is in it.

Controls & Trim

1. **Pitch, roll, yaw** — elevator (yoke fore/aft), ailerons (yoke left/right), rudder (pedals). Turns are coordinated when the ball in the turn coordinator stays centered — feet chase the ball.
2. **Trim** — the control that holds the airplane so you do not have to. In the 172, trim for the speed you want and the airplane flies itself to a large degree.
3. **Ground handling** — on the ground, the rudder steers (the 172 has a steerable nosewheel via the pedals); brakes stop and turn you tightly. The yoke does almost nothing on the ground.

Navigation & Radio

1. **Dead reckoning** — heading + speed + time = where you should be; look outside and confirm.
2. **GPS basics** — the sim's GPS/avionics show position, direct-to navigation, and a flight plan. Wave 0: fly direct and learn the moving map.
3. **Radio stack** — COM (talk), NAV (navaids), transponder (identification). Tune the airport's CTAF, listen to ATIS, and set the transponder code when ATC asks.

Weather & Environment

1. **Wind** — the day's wind is the pattern's secret boss: headwind shortens your ground legs, crosswind demands a crab on final and a sideslip at touchdown.
2. **Visibility & ceiling** — VFR weather in Wave 0: clear skies, and if the ceiling looks thin, look for the airport before you leave it.
3. **Live weather** — the sim can pull real-world weather; for training, prefer clear-and-calm until the pattern is automatic.

Wave 0: Graphics

Wave 0's graphics rule is simple: nothing here may change how the sim **behaves** — only how it looks and how readable it is. High-contrast, legible presentation is the theme; prettiness that costs frame rate is not, because Wave 0 is where you must see the instruments clearly while learning to land.

The foundational toolset already includes the two visual additions Wave 0 uses:

1. **AutoOrtho** — optional from day one: photoreal ground makes the pattern and the scenery readable, at the cost of bandwidth/cache (see the Foundational chapter).
2. **X-Camera** — your custom views; the “graphics” of how you watch yourself fly.

Beyond that, any visual mod you add belongs here and must stay cosmetic: textures, colors, HUD/interface styling, readability. Cards below are **pending addition** — per project rule, mods come from you.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Wave 1: The Ferry

Wave 1: Strategy

The Story So Far

The Ferry. Your solo is logged and your logbook has opinions. A small aircraft brokerage needs a Cirrus SR22 delivered from the coast to a buyer inland — three legs, one overnight stop, real weather, and a deadline that does not care about your nerves. You take the keys. The SR22 is the 172's confident older sibling: a glass cockpit, an autopilot, and a cruising speed that turns your home-field pattern into a memory by lunch. Your job is to plan like a professional and fly like one — cross-country navigation, honest weather decisions, and a calm hand on the controls of a faster airplane.

Strategy for the Wave

Wave 1 is where flying becomes **navigation**: the airplane is easy, the journey is not. The strategy:

1. **Plan before you fly** — every leg starts on the map (LittleNavMap is now your cockpit companion): route, altitudes, fuel stops, alternates, weather along the way.
2. **Use the tools you installed** — LittleNavMap plans and live-tracks the route; Navigraph keeps the nav data and charts current; AviTab holds the approach plates in the cockpit. These are Wave 1's superpowers — use them on every leg.
3. **Decide on weather, not hope** — check the METARs and TAFs before launch; if the destination is marginal, the alternate you planned is the professional's move. Turning around is a decision, not a defeat.
4. **Glass-cockpit discipline** — the SR22's displays give you more information than you asked for; learn what to scan and what to ignore (the mechanics chapter is the map).
5. **Autopilot as a crewmember** — let it hold altitude and heading while you manage the route and the weather; hand-fly the busy parts.

Progression Goals

1. Plan and fly a three-leg cross-country with a fuel stop and an overnight.
2. Fly a full leg in the glass cockpit using a loaded flight plan and the autopilot.
3. Make one genuine weather decision (divert, delay, or turn back) and be right.
4. Shoot your first instrument approach — even VFR, even in good weather — to learn the flow.
5. Graduate: when a 200-mile leg feels routine, the Challenger 650 in Wave 2 is waiting.

Mission Ideas

1. **The buyer's deadline** — coast to inland in one day, three legs, real live weather on.
2. **The diversion** — plan a leg into a field that goes marginal; fly the alternate you filed.
3. **The night leg** — one leg after sunset, VFR, to learn night flying's rules (visibility, horizons, and how much darker an empty sky gets).

How to Fly: The Glass-Cockpit Cross-Country

A complete cross-country leg in the SR22 G1000, told as one continuous flow rather than a checklist: the screens do the heavy lifting, and you do the thinking. Exact page labels and button names follow the aircraft's own manual — the flow is what matters, and the flow is the same on every leg of your career.

The Preflight & Plan

The leg is won before the engine starts, in LittleNavMap. Build the route — departure, destination, the altitudes for the legs, the fuel stop — and then check the weather at both ends: the METARs and TAFs, the wind at the destination, and the ceiling on the arrival. Pick the alternate while you are looking at the map, not while you are looking at the clouds. Export the plan in the sim's format and

load it into the aircraft's avionics so the map and the autopilot see exactly the route you planned, and then walk the numbers one last time: fuel on board against trip fuel plus reserve, and the plan legal before the engine turns. The preflight is the cheapest insurance in aviation, and it is the whole wave's lesson in miniature: decide on the ground, fly the decision.

Start & Program

With the plan in hand, bring the aircraft to life. Master on, then the start flow — battery, avionics, prime, start — and let the G1000 boot through its self-test while you work: the flight plan page gets the departure, the destination, and the route; the baro/altimeter is set from the local weather; the heading and altitude bugs are set for the departure. Then the check that makes the difference: the MFD's moving map should match the LittleNavMap plan leg for leg, and a mismatch found on the ground costs two minutes, while a mismatch found in the air costs a lot more. The aircraft is ready when the plan is confirmed, not when the engine is warm.

Takeoff & Climb

Line up, apply full power, and rotate at the SR22's typical speed, climbing at the book's best-rate speed with the nose pitched for the attitude. Once you are trimmed and climbing, the autopilot earns its keep: altitude select for the cruise altitude, heading select for the departure turn, and then navigation tracking once you are established on the first leg. The phrase to remember is **set and check**: the autopilot holds what you told it, so glance at the map and the groundspeed to confirm you are actually going where the plan thinks you are. The first leg is where the plan becomes a fact, and the check is what makes it yours.

Cruise on the Autopilot

The cruise is where a cross-country leg becomes a study in boredom, and boredom is the point. Level at the cruise altitude, manage power and mixture for the book's cruise settings, and know your numbers — the displays show fuel flow and endurance, and the numbers you know before you need them are the numbers that keep you safe. Watch the weather picture as it evolves; if the destination sours, the alternate from the preflight is already known, and the decision is made early or not at all. And use the sim's gift: time warp. Speed up the long, straight cruise and slow down for the work — but remember where you are when you come back, because the leg you are flying is the one you are responsible for.

Descent & Arrival

The arrival is the flight's second decision point, and it is made before the airport appears. Plan the descent so you arrive at the destination's pattern or approach altitude in time: the altitude to lose, divided by your descent rate, plus a margin — the math is simple, and it is what keeps an arrival from becoming a rescue. Brief the approach from the plate in AviTab — runway, altitude, frequency, minimums — out loud, because the habit is the skill. Then configure: power back, flaps as the flow demands, speed for the approach, and let the autopilot or your hand fly the final course while you manage the energy. The descent is the moment the whole flight plan converges on the runway, and the pilot who briefed it is the pilot who is calm.

Approach & Landing

The landing is the same one you learned in Wave 0, in a faster airplane with more on the line. Line up the aim point — the runway threshold fixed in the windshield — and adjust power for the glide and pitch for the speed; the round-out over the threshold is the same gentle flare, and the mains settle the same way. A crosswind leg means a crab on final and a kick at the flare, and it is the SR22's only secret: practice it, because the ferry routes you fly in this wave will give you plenty of crosswinds. Rollout, brakes to taxi speed, exit at the taxiway, and the leg is yours.

The Debrief

Every leg deserves ten minutes of review. Replay the landing in the sim and watch what your hands actually did — a rough touchdown almost always has a visible cause: speed, flare, or staring at the runway instead of the horizon. Check the track against the plan in LittleNavMap: where did you drift, and what would you change? Then log the leg, because every leg is a data point, and Wave 2's instrument work is built on the habit of reviewing what you flew. The ferry pilot's logbook fills up with reviews, not just flights, and that is what makes the next leg better.

Wave 1: Modlist

Wave 1 keeps Wave 0's quality-of-life base (the foundational toolset) and begins adding **content and mechanics** — but no pure power spikes. The rule for this wave: anything that adds capability must also add something to learn or manage.

The wave's centerpiece is a **user-supplied aircraft addon**: the Cirrus SR22 G1000 in the form of the Take Command SR22 G1000 Series you supplied.

SR22 G1000 Series (Take Command)

Dependencies: user to provide

Impact: Adds the Cirrus SR22 G1000 as this wave's study aircraft — the glass-cockpit platform the wave guide is written around.

Notes: User-supplied aircraft (payware addon). Current version and any dependencies: user to provide. Per project rule, nothing here validates compatibility — confirm against 12.4.3 with the release-date heuristic.

Per project rule, further wave-specific mods come from you — cards below are **pending addition**, with the fields this guide's rules demand. Anything you do not supply stays “**user to provide**”.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Wave 1: Mechanics

The SR22 is the classroom for the systems every later wave assumes: the glass cockpit, the autopilot, and the planning workflow that turns a flight into a route. None of it is difficult — the difficulty is in the discipline, and this chapter is about the discipline.

The Glass Cockpit

The SR22's panel replaces the 172's round gauges with two large screens. The primary flight display (PFD) puts airspeed, attitude, altitude, heading, and vertical speed together in one place, with the autopilot's targets displayed alongside so you can cross-check what you asked the aircraft to do against what it is actually doing. The multi-function display (MFD) is the map: the moving map draws your position on the route, and with a flight plan loaded it shows the waypoints, the legs, and the procedures ahead. Some installations add synthetic vision and terrain, where the screen draws the ground ahead of you with terrain and obstacles highlighted, so the map becomes a window.

The glass changes the instrument scan in a subtle way: all the numbers are in one place, but the scan still matters, because the danger of a big display is fixation. The classic scan still applies — attitude, airspeed, altitude, heading, with the map for the big picture — and the skill is in sweeping across the

displays rather than staring at the prettiest one. Learn to read the glass the way you read the round gauges: as a set of instruments you move between, not a movie you watch.

Autopilot Basics

The autopilot is a crewmember, not a replacement. The SR22's autopilot holds the modes you select: heading select turns when you tell it to and holds the course; altitude select climbs or descends to the target and holds it; vertical speed manages the rate; and navigation tracking follows the loaded route through the GPS. Learn them in the order of use — heading and altitude first, navigation tracking once the route is in the box — and they become the same ideas you will manage at airline scale in the 737.

The discipline has two rules. First, engage with the aircraft trimmed and stable: the autopilot flies what it is given, and handing it an untrimmed, unstable aircraft is handing it a problem. Second, disengage by hand before you need to — the autopilot should never fight you, and the pilot who reaches for the disconnect at the same moment as the problem is the pilot who is already behind. The autopilot reduces workload; it does not remove it. You still manage power, configure the aircraft, and watch the weather, and the moment you stop doing those because “the autopilot is flying” is the moment you are no longer the pilot.

Flight Planning Workflow

Wave 1 is where flying becomes navigation, and navigation is a workflow that starts before the engine. Build the route in LittleNavMap — departure, destination, route altitudes, fuel stops, alternates — and check the weather along the way, deciding that the flight is legal and sane: VFR minimums at both ends, fuel with margin, and an alternate you could actually reach. Export the plan in the sim's format and load it into the aircraft's avionics, so the map and the autopilot see the route you planned. Then fly it: the moving map and the autopilot fly the plan while you manage the aircraft and the decisions. And review it after landing, because LittleNavMap keeps the track and the debrief — where you drifted, what you would change — is where the learning lives.

The workflow is the same skeleton every later wave uses, with more elaborate tools bolted on. Learn it now, simply, and the FMS of Waves 2-5 is just this workflow with a bigger screen.

Navigation & Procedures

The SR22's avionics include the classic nav aids alongside the GPS, and the fundamentals matter even in the age of the moving map. VOR navigation is a needle and a course: tune the station, set the course, and fly the needle, and the skill transfers directly to the ILS, where the localizer and glideslope needles are the same idea guiding you to the runway. The approach flow is worth learning before you need it: brief the plate in AviTab, fly to the initial fix, join the final course, and decide early on a go-around if it does not feel right — the decision that feels late in the pattern is the decision that is late in the weather.

Radio work grows up in this wave. Read back clearances, announce positions at non-towered fields, and let the sim's ATC (or a busy frequency) make you practice out loud: the phraseology is a language, and the only way to be fluent is to speak it.

Weather Decision-Making

The professional's relationship with weather is numbers, not feelings. A METAR and a TAF tell you the wind, the visibility, the ceiling, and the trend — the numbers that make or break a VFR flight — and reading them before every leg is the habit that keeps a ferry pilot employed. The destination is only the destination if you can also leave it: fuel and alternates are part of the plan, not an afterthought, and the day's route should always know where Plan B is.

The go/no-go call is the real curriculum. It is easier to explain a cancelled flight than a forced landing, and the professional's rule is that the call gets made with a clear head on the ground, never with the

fuel low and the ceiling dropping. Wave 1's real test is not the flying; it is making the call, calmly, and being right.

Wave 1: Content

Content mods for this wave are the places, not the systems: scenery along the ferry route that makes cross-country flying feel real — airports, landmarks, regions, and anything that adds a reason to fly a particular leg.

Per project rule, this project does not select mods — the cards below are **pending addition** from you, with the fields this guide's rules demand.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Wave 1: Graphics

Wave 1 is where the world you fly over starts to matter — long legs over real terrain deserve to look real. Visual mods here are allowed as long as they stay cosmetic.

The foundational toolset already covers the big one: **AutoOrtho** (photoreal ground, see the Foundational chapter) is at home in this wave's cross-country legs.

Per project rule, additional visual mods come from you — cards below are **pending addition**.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Wave 2: The Charter

Wave 2: Strategy

The Story So Far

The Charter. The ferry flights are logged, and a two-man executive charter outfit has noticed. Their **Challenger 650** — a big-cabin business jet with a flight deck of screens and systems depth that would make a regional airline blush — flies clients to places the schedule never goes: a factory in the mountains, a meeting across an ocean, an FBO at midnight. They need a captain who can fly **in** the system, not just around it: turboprop power, a pressurised cabin, a full FMS, and instrument approaches in real weather. You take the left seat. The pattern is over; the flight plan is now the default state of your life, and the clients are the reason every leg has to be perfect.

Strategy for the Wave

Wave 2 is the **IFR** wave: from here on, flying means flying the system — clearances, approaches, minimums, and the discipline to fly instruments when the world goes white.

1. **Jet transition** — the Challenger's turboprops are a different animal from the SR22's piston: thrust is managed by the fans' speeds (N1/N2) and temperature limits (EGT), not by torque and mixture, and the automation does much of the engine management for you. Learn the numbers before you need them (mechanics chapter).
2. **IFR by the book** — fly every leg IFR: filed plan, ATC in the loop (the sim's ATC), SID/STAR where they exist, and an instrument approach to minimums when the weather asks for it.
3. **The FMS is your copilot** — the flight management system flies the route, the approaches, and the vertical profile. Program it, cross-check it, and let it work while you manage the airplane and the decisions.
4. **Workload management** — single-pilot IFR in a business jet is a course in triage: aviate, navigate, communicate — in that order, always. When it gets busy, fly the airplane first.
5. **Weather is a system** — fronts, icing, and the minimums on the plate are inputs to a decision, not drama. The professional's calm is the point of this wave.

Progression Goals

1. Complete a cold-and-dark start of the Challenger 650 using the aircraft's checklist flow.
2. Fly a full IFR leg: filed plan, clearance, SID/STAR, FMS route, approach to landing.
3. Shoot an ILS approach to minimums in genuinely marginal weather and land.
4. Handle one emergency procedure from the aircraft's checklist (engine, pressurization, or icing scenario) without drama.
5. Graduate: when an IFR leg is routine, the Q400's regional grid in Wave 3 is your next commute.

Mission Ideas

1. **The urgent delivery** — a cargo charter into a field with a marginal forecast; decide, file, and fly it (or divert like a professional).
2. **The night IFR run** — a night leg in weather: the full system, no sightseeing.
3. **The icing escape** — plan a leg that meets an icing layer; use the aircraft's anti-ice and the weather picture to get out of it cleanly.

Learning IFR

IFR is not a different way to fly — it is the same flying, done when you cannot see, which changes everything about how you think. Wave 2's job is to make the instrument world feel normal: the scan, the procedures, and the calm of an approach flown to minimums in the murk. The Challenger 650 is the classroom, but everything here transfers to every instrument panel you will ever sit behind.

The Instrument Scan

The foundation of all instrument flying is the scan: a deliberate, continuous pattern of reading the instruments in a fixed order so that nothing gets ignored when it matters. The order is not arbitrary. **Attitude** is the anchor — the artificial horizon tells you pitch and bank directly, and it is the one instrument that cannot lie to you about what the aircraft is doing. Everything else confirms or refines it: **airspeed** tells you whether the attitude you set is producing the performance you want; **altitude** is the contract with ATC and the terrain; **heading** is where the aircraft points; **vertical speed** is the trend, a moment behind the others, so it confirms rather than leads; and **the needle** — localizer, glideslope, or course — is the path itself.

The classic way to practice is to fly a plain, boring leg in clear weather with the scan running the whole time: pick an altitude, hold it, hold a heading, and keep the scan moving — a few seconds per instrument, never staring at any one. The day it matters — ceiling at minimums, no horizon, an approach to fly — the scan is already a habit, and habits do not need thinking.

The Cold & Dark Start

Starting the Challenger from cold and dark is the wave's ritual, and it teaches the aircraft's systems in the most natural way possible: one by one, in the order they need each other. Begin with the cockpit setup — avionics on, the FMS booted, and the flight plan loaded and cross-checked against the LittleNavMap plan you built on the ground; a mismatch is far cheaper to find with the aircraft parked. Then the start flow, per the aircraft's checklist: fuel on, the start sequence, and the temperature limits watched like a hawk — the first start of the day is where hot starts happen. Finish by programming the day into the boxes: performance numbers from the FMS, the departure procedure loaded, the altimeter set. When the aircraft is ready before the passengers are, the start was good.

The IFR Leg Flow

Every IFR leg is the same skeleton, and flying the skeleton in the same order every time is what makes the work feel manageable. It starts with the clearance — read it back, load it, fly it: “cleared as filed” or the full route, the SID if there is one, the altitude. The departure then takes you up the procedure as published: altitude and course changes at the fixes, not when you feel like it. In the climb and cruise, ATC (the sim's, in our case) steps you up, the autopilot holds the mode you selected, and you manage the radio and the fuel. The descent is where the flight is actually won: the STAR if there is one, then vectors or the arrival — and the approach is **briefed before you reach it**, not after, because an unbriefed approach is the instrument pilot's version of flying blind. Finally the approach itself: armed, captured, and flown to minimums, with the missed approach already decided. Same skeleton every leg, and the skeleton is the skill.

The Approach Briefing

The briefing is the approach done in miniature, on the ground or in the cruise, from the plate in AviTab. Work through it in the same order every time: the **setup** — runway, frequency, the navaid or RNAV waypoints, and the initial approach fix; the **path** — courses, altitudes, and the final approach course, including where the glideslope or lateral capture should happen; the **minimums** — decision altitude and visibility, and what you must see to continue; the **missed approach** — where you climb, what you turn to, and what you fly next; and the **go/no-go** — the weather at the field and the fuel to the alternate. Brief it the same way a hundred times and the hundred-and-first — in real weather, at the end of a long day — briefs itself.

Flying the ILS

The ILS is the classic instrument approach, and flying it well is a study in patience. Intercept the localizer — the lateral needle — well before the final approach fix, at the altitude the plate specifies, and let the needle come to you rather than chasing it; an overshoot is almost always the result of

trying to catch a needle that was already converging. When the glideslope centers, capture it: pitch for the path, power for the speed, and make small corrections — the instruments reward the pilot who nudges, not the one who grabs. Crosswind complicates it honestly: you crab into the wind to hold the localizer, and the wind correction is part of the approach, not an error.

At the decision altitude the briefing pays off: if the runway environment is in sight, continue to a normal landing; if it is not, the missed approach is not a failure — it is the procedure you briefed two hundred miles ago. Hand-fly the first attempts, even if the autopilot can do it, because the autopilot will not be there the day the needles matter most. The ILS is where the scan becomes a dance; let it.

The Missed Approach

The missed approach is part of the approach, not an emergency, and the distinction is what makes it calm. When you call the go-around — at minimums without the runway, or any time the approach is not working — the sequence is fixed: power and pitch, climb thrust and nose to the missed altitude; clean up, with the flaps retracting on schedule as speed builds; then fly the plate, the published missed procedure with its climb, its turn, and its hold or its next approach. And then decide: hold, try again, or go to the alternate — a decision that was already made on the ground, because the fuel and the plan were built around it. Fly the missed approach often enough that it is boring, and it will never be frightening.

The Practice Progression

Learn IFR the way you learned the pattern: in layers. First, run the scan on clear-weather legs until it is a reflex — fly a leg at a set altitude and heading and make the instruments your scenery. Second, fly one complete IFR leg a day, clearance to touchdown, all of it, even in perfect weather — the skeleton becomes muscle memory. Third, fly it in weather: real minimums, real decisions, and the quiet satisfaction of an approach that works when it has to. Fourth, shoot the same approach twice — once with the autopilot flying it, once by hand — and compare: the autopilot shows you what the aircraft can do, the hand-fly shows you what you can do. When the whole flow feels like routine, the Q400 in Wave 3 puts it to work at regional pace, where the routine is the schedule.

Wave 2: Modlist

Wave 2 builds on the foundational toolset and Wave 1's additions, focused on the IFR systems: navigation data, charts, and planning tools are now **assumed** — the foundational Navigraph and LittleNavMap are central to this wave.

The wave's centerpiece is a **user-supplied aircraft addon**: the Challenger 650 in the form of the Hot Start Challenger 650 you supplied.

Hot Start Challenger 650

Dependencies: user to provide

Impact: Adds the Challenger 650 big-cabin business jet as this wave's study aircraft — the aircraft the wave guide is written around.

Notes: User-supplied aircraft (payware addon). Current version and any dependencies: user to provide. Per project rule, nothing here validates compatibility — confirm against 12.4.3 with the release-date heuristic.

Per project rule, further wave-specific mods come from you — cards below are **pending addition**, with the fields this guide's rules demand. Anything you do not supply stays “**user to provide**”.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Wave 2: Mechanics

The Challenger 650 is the aircraft that teaches you the systems every professional turbine pilot works with for the rest of their career: turbofan engines that largely manage themselves, a cabin that flies at a different altitude than you do, an FMS that does most of the navigation, and the instrument discipline that makes all of it safe in the weather. None of it is hard on its own — the skill is in how the pieces fit together, and that is what this chapter is for.

Turbofan Power

The engines under the Challenger's wings are a different species from the SR22's piston. A turbofan breathes: air enters the big fan at the front, and most of it bypasses the core to produce thrust directly, while the rest feeds the core compressor, the combustor, and the turbines that drive both the fan and the core. That is why the two speeds on the engine display matter. N1 is the fan speed, and in practice it is your **thrust gauge**: in almost every regime you fly, "power" means an N1 target, not a lever position. N2 is the core speed — the "engine is alive and healthy" number: it spins up with the starter, settles into a stable idle, and tracks its own limits as you fly.

What makes a turbofan feel different to fly is how it responds to your hand. A piston engine answers instantly; a turbofan **spools**. Push the levers forward and the thrust builds over a couple of seconds; pull them back and it decays slowly. Every power change in the Challenger has to be anticipated rather than reacted to — on the approach you set thrust for where the aircraft will be five seconds from now, not where it is now. The auto-throttle (where your installation has it) smooths much of this for you by holding an N1 target, but the habit of thinking ahead of the spool is yours to build, because it is the same habit you will need in the Q400 and the 737.

The third number on the engine page is EGT, exhaust gas temperature, and it is the engine's limit in human terms. EGT climbs with power, with heat, and with slow speed — a hot day, a heavy aircraft, and a shallow climb all push it up — and the rule is simple: never hold it in the red. Watch EGT the way you watched the stall warning in Wave 0: not the primary instrument, but the one that tells you when you are asking too much of the machine.

Pressurization & Cabin

The cabin flies at a different altitude than you do, and that is a system you manage. Bleed air tapped from the engines pressurizes the fuselage; the pressurization controller lets you set the **cabin altitude** you want, and it then climbs and descends the cabin at a gentle, comfortable rate while the aircraft does its own, faster thing. The client in the back seat feels the cabin, not the aircraft — so the rule of the charter pilot is that cabin rate limits your climb and descent, not the flight path. Set the cabin altitude target for the cruise altitude before you climb, and the controller does the rest.

The same system makes the descent a planning exercise. A turbine aircraft can come down very fast; the cabin cannot. A good arrival starts with a top-of-descent calculated so the cabin pressure and the flight path arrive at the airport together — begin down too late and you either rush the cabin down or float high and fast at the runway. It is the first hint of the energy management that becomes the whole game in later waves: plan the descent, then fly the plan.

FMS & Navigation

The Challenger's flight deck — a Pro Line 21-style suite of large displays and a full flight management system — is the cockpit of the rest of your career. The FMS is a route computer: you load a plan into it (the same plan you built in LittleNavMap), add the departure and arrival procedures, and it flies

the lateral and vertical path, turn by turn and step by step, while you manage the aircraft and the decisions around it. The important mental model is that the FMS is **your** tool, not a substitute for you: it executes exactly what you entered, so entering it carefully and cross-checking it are the skills that matter.

The approaches you fly with it come in two families. RNAV/GPS approaches let the box guide you along a defined path with vertical guidance — you watch the box do the navigating while you manage speed and configuration. ILS approaches give you the classic needles: a localizer for lateral alignment and a glideslope for the vertical path, captured and followed by hand or by the flight director. Fly both until the difference is instinct — a pilot who can only follow one is a pilot who is one weather system away from being stuck.

IFR Discipline

Instrument flying is a set of habits, and Wave 2 is where they become permanent. The first is the scan: a constant, deliberate sweep of attitude, airspeed, altitude, heading, and vertical speed, cross-checked against the approach plate's numbers. The second is the culture of minimums — every approach has a decision altitude and a visibility minimum, and below them the go-around is not a choice, it is the procedure; the professional's pride is in flying the procedure, not in "making" the approach. The third is radio discipline: read back clearances, report positions, and let the sim's ATC keep you honest — the quiet confidence of someone who files and flies the system is an instrument in itself.

And the fourth habit is the weather itself. Anti-ice and de-icing equipment (engine heat, wing heat) buys you the **escape** from icing, never the right to push through it; ice is a leave condition, and the plan should include the way out before you fly into the cloud.

Workload Management

Single-pilot IFR in a business jet is a course in triage, and the triage order never changes: **aviate, navigate, communicate**. When the workload spikes — a busy arrival, a weather decision, a system caution — the airplane and the path come first, and the radio waits. Flows and checklists support this: you do the flow from memory (the same sequence every time, which is what makes it memory), then confirm against the list. Delegate to the systems that earn it — the autopilot holds altitude, the FMS flies the path — and keep the decisions that are yours: weather, fuel, minimums, and the go-around. The aircraft flies the flight; you fly the operation.

Wave 2: Content

Content for the charter wave: the network of small fields and executive destinations the Challenger 650 serves, and any scenery that makes the IFR world feel real — approach-friendly airports, region packs, and the places a charter captain actually flies into.

Per project rule, content mods come from you — cards below are **pending addition**.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Wave 2: Graphics

Visual additions this wave should serve the IFR world: weather presentation, lighting for night IFR, and readability of the instruments and plates you fly by. The foundational toolset's AutoOrtho and X-Camera remain in play.

Per project rule, visual mods come from you — cards below are **pending addition**.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Wave 3: The Dash Ace

Wave 3: Strategy

The Story So Far

The Dash Ace. The charter years made you a systems pilot, and a regional airline likes the look of that. The schedule you come to love is the one the big jets cannot fly: the regional grid — short legs, tight turns, small fields, and a turboprop that cruises like a jet and lands like a mountain goat. The airline hands you the left seat of the Dash 8 Q400, the fastest turboprop airliner in the world, and a timetable measured in minutes, not hours. The Q400 is a pilot's airplane: turbine power, a highly automated cockpit, and performance demands (steep approaches, precise energy management, prop handling at the edges) that keep you honest every single leg. This is where **ace** stops being a title and becomes a skill set.

Strategy for the Wave

Wave 3 is **high-performance turboprop operations**: jet speeds, prop rules, and the brutal efficiency of a regional schedule.

1. **Fly the prop, think the jet** — the Q400 cruises at airliner speeds, but the engines are turbo-props: torque, temperature limits, and propeller management (beta, reverse, feather) are the daily mechanics of every flight.
2. **Energy is everything** — steep approaches and short runways are the Q400's signature: the descent is a continuous energy calculation, and the propeller's reverse/beta authority is your brake. Fly the profile; do not chase it.
3. **The minute game** — regional flying lives and dies on turnaround: brief quickly, program the FMC fast, and fly a clean first time. The Q400 rewards the prepared and punishes the reactive.
4. **Weather at low altitude** — the regional grid lives in the weather the big jets climb out of: icing, wind shear, and marginal ceilings are the daily commute. Your Wave 2 instrument discipline and the Challenger's automation habits now run at regional pace.
5. **Aircraft addon** — this wave's aircraft is the **FlyJSim Q4XP**, a payware Dash 8 Q400 you supplied as the wave's study aircraft (see the Modlist section for the card and its status).

Progression Goals

1. Master the Q400's flows: cold start, FMC program, and a full regional leg with callouts.
2. Fly a steep approach and short-field landing using the propeller's beta/reverse authority precisely.
3. Run a three-leg regional day on time, in weather, with an alternate at every stop.
4. Handle an engine-out drill and an icing encounter per the aircraft's checklist without drama.
5. Graduate: when the minute game feels slow, the 737 line in Wave 4 is your next seat.

Mission Ideas

1. **The grid day** — a morning bank of three short legs with 20-minute turns and weather at every field.
2. **The island run** — a short, steep approach into a field that barely has a runway; the Q400's specialty.
3. **The icing morning** — a cold day with freezing levels right at pattern altitude: use the de-icing systems and the weather picture to run the schedule without bending it.

How to Fly: The Regional Turboprop

The Q400 day, step by step — the flows, the leg, and the steep approach that makes the type famous. The aircraft's own manual has the exact numbers; this chapter teaches the rhythm, and the rhythm is what turns a busy regional schedule into a comfortable day.

The Q400 Flows

The Q400 is flown in flows — a fixed sequence of checks done the same way every single time, until the sequence is memory and the hands do it without being told. It starts with the cockpit setup: avionics on, the FMS booted, the plan loaded from LittleNavMap and cross-checked leg by leg, and the day's performance numbers in the boxes. Then the start flow: engines one by one, the temperature limits watched on each start, and the props brought up to governed rpm once the engines are stable. Then the taxi flow: brakes, trim, and the pre-takeoff checks in the same order every time. None of it is hard; the point of a flow is that nothing gets skipped when the schedule is pressing, because the schedule always presses.

The Regional Leg

A regional leg is a short, dense story with three acts. The first is the brief, done in two minutes on the ground: weather, runway, the departure, the fuel — the whole day condensed into what you need for this leg, so the cockpit work is confirmation rather than discovery. The second is the departure and climb: the SID or the runway heading, up to the leg's altitude, and the autopilot takes the route from the FMS as soon as you are clear. The third is the arrival: the STAR or the vectors, the approach briefed from the plate in AviTab, and the energy managed from the top of descent — the Q400 does not cruise, it commutes, and the next approach should be in your head from the moment you level off. Land, turn around, and the next leg's story begins: brief, fly, arrive. The rhythm is the schedule.

The Steep Approach

The steep approach is the Q400's signature, and it is a technique rather than a fear. Fly the published approach gradient as a constant-rate descent with power: the aim point stays fixed in the windshield — not sliding up, not sliding down — and the rate stays constant, with power doing the fine tuning. The energy account is the whole game: too high, and the fix (power back) is easy early and hard late; too low, and the runway gets small while the options get few. The Q400 rewards deciding early, and the way to learn that is to practice the profile on a long runway first, where a mistake is a learning moment rather than a go-around, and then take the same approach to a short field, where it is just the same approach with a better excuse.

The Beta/Reverse Landing

The landing itself is where the propeller stops being a propeller. Touch down at the right speed and the flare is short and gentle — the airplane lands like it was always going to. Then the props go to beta: the blades flatten and the propeller becomes a precise deceleration control, letting you steer the stop with power rather than just the brakes. Where the runway demands it — and a short regional strip often does — reverse follows: the blades push air forward and the aircraft stops in a hurry. Use reverse like the brake it is, smoothly and deliberately, not like a party trick: touchdown point, beta, reverse, brakes, in that order, and the Q400 stops where you meant to stop, every time.

Tip: Practice the steep approach on a long runway first: it lets you learn the profile without the pressure of the short field. The short runway is then just the same approach with a better excuse.

The Go-Around

In a high-drag, high-power airplane, the missed approach is an energy decision, and it is a normal part of the operation. Call it early — at any point the approach stops working — and the sequence is fixed: power and pitch, climb thrust and nose to the missed approach speed; clean up on schedule, with the flaps retracting in the flow rather than all at once; and then fly the missed procedure, the published climb and turn, with the regional day continuing on the other side. Brief it like the approach, fly it like the approach, and log it like the approach: the go-around is not a failure of the landing, it is a success of the judgment.

Wave 3: Modlist

Wave 3's centerpiece is a **user-supplied aircraft addon**: the Dash 8 Q400 in the form of the FlyJSim Q4XP you supplied. Everything else in this wave rides on the foundational toolset and the systems added in Waves 1-2.

FlyJSim Q4XP (Dash 8 Q400)

Dependencies: user to provide

Impact: Adds the Dash 8 Q400 turboprop airliner as this wave's study aircraft — the aircraft the wave guide is written around.

Notes: User-supplied aircraft (payware addon). Current version and any dependencies: user to provide. Per project rule, nothing here validates compatibility — confirm against 12.4.3 with the release-date heuristic.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Wave 3: Mechanics

The Dash 8 Q400 is a jet pretending to be a propeller aircraft. It cruises at airliner speeds, carries a highly automated cockpit, and yet the power comes from two large turboprops whose propellers are as much brakes as thrust. That combination — jet speeds with prop rules — is what makes it a pilot's airplane, and this chapter is about the systems that make that combination work.

The High-Speed Turboprop

After the Challenger's turbofans, the Q400 feels like a step back to the King Air's world — torque and temperature, the language of propellers — but with far more of everything. Power is torque: the engines' twisting force on the propeller shaft, displayed as a percentage, and the number you manage with the power levers. The law is temperature, exactly as it was on the Challenger but with a different gauge: watch the turbine temperature limits on hot days and at high power, because a turboprop's turbine does not forgive an exceedance any more than a turbofan's does.

What the Q400 adds is the propeller as a control surface. In flight the props run at their governed rpm and behave like thrust — but at the edges of the flight envelope they do the things only props can do. **Beta** is the pitch range below flight idle: the blades flatten toward the windmill position and the propeller becomes a precise brake and steering device for ground operations. **Reverse** takes the blades past flat pitch so the propeller pushes air **forward**, and the aircraft stops with authority — this is the Q400's real braking, used on short runways the way a jet uses thrust reversers. And **feather** is the emergency pitch: if an engine quits, its propeller is feathered edge-on to the airflow to stop it windmilling and dragging. The Q400's asymmetric-thrust behavior with one engine feathered is dramatic and physical — the rudder does real work — and the drill belongs in your muscle memory from the first week, practiced at altitude with a simulated failure until it is calm.

Steep Approaches & Energy

The Q400's steep approach is not a stunt; it is a procedure born of noise rules and short fields, and it teaches the skill every pilot needs forever: energy management. The airplane descends on a constant, steep gradient with power — the aim point stays fixed in the windshield, the rate stays constant, and power does the fine tuning. The mental model is an energy account: too high and you have too much

energy (fix it early with power, or the runway gets small); too low and you have too little (fix it with power and pitch before the runway becomes the problem). High is almost always fixable; low is not, and the Q400 rewards the pilot who decides early.

The short-field landing is the steep approach's payoff. Plan the touchdown point, use the propeller's beta and reverse authority, and stop where you meant to stop — a landing that looks impossible from outside and feels routine from inside. It is the party trick of the type, and it is entirely a matter of flying the profile rather than chasing it.

The Automated Cockpit

The Q400's cockpit automates the route the way the Challenger's did, at regional scale: the FMS holds the plan — exported from LittleNavMap, as always — and flies the lateral and vertical path while you manage the aircraft and the day. The difference is pace. A regional schedule lives and dies on the turnaround: brief quickly, program the FMS fast, taxi without rework — the aircraft punishes the unprepared and rewards the prepared. The two-pilot discipline you built in the charter wave — flows, checklists, and callouts on every leg, even solo — is what makes the pace survivable: say the callouts, do the flow from memory, confirm with the list, and the minute game becomes routine rather than chaos.

Regional Weather & Icing

The regional grid lives in the weather the big jets climb out of, and the Q400 spends its working life there. Icing is the daily conversation: the aircraft's de-icing systems — boots and heat — are effective, but they buy the escape, not the right to push through; ice is a leave condition, and the route should have an out before the cloud is entered. Wind shear is the other low-altitude hazard: treat every marginal day as a decision day, brief the approach with the plate and the live picture, and fly the go-around without drama if the approach stops working. And the regional habit that makes all of it safe is the alternate: short legs, quick alternates, and fuel and a plan that always know where Plan B is.

Wave 3: Content

Content for the regional wave: the grid — small hubs, short fields, and the route network the Q400 serves. Scenery that makes a fast-turnaround regional day feel real belongs here.

Per project rule, content mods come from you — cards below are **pending addition**.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Wave 3: Graphics

Visual needs this wave: weather presentation that makes the regional commute real, and lighting for the early-morning/late-evening grid. The foundational toolset (AutoOrtho, X-Camera) stays in play. Per project rule, visual mods come from you — cards below are **pending addition**.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Wave 4: The Line

Wave 4: Strategy

The Story So Far

The Line. The regional years proved you could run a schedule, and mainline noticed. Now you sit in the right seat of a Boeing 737-800 — and within a year, the left. The schedule is a spreadsheet; the airplane is a system; the passengers are people who think “today” means “now”. Your world has changed scale: a single leg is three hundred souls and a hundred tonnes of jet at thirty-five thousand feet, and the margin for casual error is exactly zero. This wave is where the sim stops being a flight game and becomes an operations job — and where the **ace** in Ace’s Realm starts to mean something.

Strategy for the Wave

Wave 4 is **airline operations**: the aircraft is a well-behaved machine that will fly itself perfectly — your job is the operation around it.

1. **The FMC is the contract** — the flight management computer holds the route, the performance, and the vertical profile. Program it completely, cross-check it twice, and fly what it says while it flies what you programmed.
2. **Crew discipline** — flows and checklists, callouts and readbacks, and the sterile cockpit: below ten thousand feet, the talk is flying. Even in single-pilot practice, **say** the callouts — the habit is the safety.
3. **Plan the whole day** — fuel, payload, alternates, and the weather at both ends: the airline plan starts before the pushback, in the briefing, not in the cockpit.
4. **Automation as a tool** — the autopilot and auto-throttle fly; you manage the automation: modes, targets, and the constant question of whether the box is doing what you think it is.
5. **The descent is the flight** — most of the challenge is getting from cruise to the runway: the top-of-descent decision, the arrival, the approach, and the go-around that is always legal and always available.

Progression Goals

1. Complete a full airline leg: briefing, cold-and-dark start, FMC setup, pushback (Better Pushback), taxi, takeoff, cruise, descent, ILS approach, landing, and shutdown — with callouts throughout.
2. Fly a departure and arrival with published SID/STAR procedures in the FMC.
3. Manage a fuel/performance scenario (alternate, diversion, or holding) from the briefing data.
4. Fly one approach to autopilot-guided minimums and hand-fly the landing from the flare — the transition that separates simmers from operators.
5. Graduate: when the line feels like routine, the 777F in Wave 5 is the biggest desk in the company.

Mission Ideas

1. **The morning bank** — three legs in one day with real weather and a connection to protect.
2. **The diversion** — the classic airline drama: destination goes below minimums; fly the alternate and explain it with the calm of someone who planned for it.
3. **The night edge** — a late arrival into a busy airport with the arrival, the sequence, and fatigue (your own) as the story.

The FMC, Step by Step

The Flight Management Computer is the 737’s cockpit brain: route, performance, and vertical profile in one box. Learn the flow once, and every FMC in this guide — the Q400’s, the 777’s — is the same idea in different clothes: a set of pages you work through in a fixed order, entering data and cross-

checking what the box thinks it knows. Exact page names follow the aircraft's manual; the order is universal, and the order is the skill.

The Pages

An FMC is a filing cabinet of pages, and the work flow is the same every flight. You begin with the identity and initial pages — who the aircraft is and where it is. Then the route: the flight plan, waypoint by waypoint, with the departure and arrival procedures added from the DEP/ARR pages. Then the performance pages, where the numbers live: weights, fuel, speeds, and the takeoff data the box computes from the runway and the weather. Then the LEGS page, the route in detail — each leg, its course, its distance, and the vertical constraints — which is the page you check twice, because it is where the box shows you exactly what it thinks it is going to fly.

Load the Plan

The plan lives first outside the box, in LittleNavMap, built and checked while the aircraft is parked: route, altitudes, the SID and STAR you expect, and the approach. Enter it into the FMC — waypoint by waypoint, or from a saved plan where the aircraft supports it — and watch the box build the legs as you go; the box should confirm the plan, never invent it. Add the procedures: DEP for the departure, ARR for the arrival and the approach, and the FMC inserts the published fixes and the vertical profile. Then cross-check the LEGS page against the briefing, leg by leg: distances sane, altitudes legal, nothing doubled. It is the step that catches the mistakes, and it is the step the hurried pilot skips.

Performance & Takeoff

With the route in, the performance pages turn the day into numbers: the zero fuel weight, the fuel load, and the V-speeds the box computes — V₁, V_r, V₂ — for the runway and the weather you entered. Arm the thrust and the auto-throttle for the departure, and brief the takeoff out loud: the speeds, the departure, the abort plan. The box is now carrying everything it needs; the takeoff itself is the part you fly by hand and by the numbers, with the box as the source of those numbers.

Fly the Path

After takeoff, the box starts earning its keep. LNAV — lateral navigation — flies the route turn by turn: engage it when the departure allows, and watch the first turn, because the first turn tells you whether the box has the plan you think it has. VNAV — vertical navigation — flies the profile: the climb at the computed speeds, the level-offs at the constraints, and the descent beginning at the computed top of descent. The box only knows what you entered, so the constraints on the LEGS page are the contract, and the descent is the moment of truth: if it looks early or late, the energy is the lever — speedbrakes and speed, not panic — and a well-programmed box makes the arrival a quiet confirmation rather than a rescue.

The Arrival & Approach

The arrival and approach are the box's last act, and yours is the briefing. As the descent begins, load the arrival and arm the approach: the ILS frequency or the RNAV approach, the minimums set, the final approach course in the box. Watch the captures — the localizer, then the glideslope — and let the box fly the approach while you fly the briefing: the decision altitude, the runway environment, and the call. The box does the flying; the pilot does the deciding, and the deciding is the job.

The Autoland Brief

Some 737 approaches can go all the way to touchdown on the automation, and an autoland deserves the same briefing as any approach: the setup, the path, the minimums, the missed — plus the autoland specifics, the dual autopilots engaged where the aircraft requires it, the flare armed. Then monitor like a hawk: the needles, the speed, and the mode annunciations, because the automation is a tool, not a witness. At minimums the decision is still yours: continue to the automatic flare, or take it by

hand. The point of the autoland brief is not the novelty of hands-off landing; it is that even the most automated approach in the airplane belongs to the same discipline as the first one you briefed in Wave 2.

The Cross-Check Habit

The FMC is only as good as its data, and its data is only as good as your entry — so the habit is a rhythm of checks, each cheap and each load-bearing: once on the ground, the route you loaded matches the plan you briefed; once in the climb, the LEGS page still matches the clearance; once before the approach, the loaded approach is the briefed approach. The box gets no free pass, ever. That is the airline way, and it is the point of Wave 4: the automation is powerful, and the pilot who treats it as a proposal to be verified, every time, is the pilot the automation can never surprise.

Wave 4: Modlist

Wave 4 assumes the full foundational toolset — Navigraph (current nav data and charts are now load-bearing), LittleNavMap (briefing and planning), AviTab (plates in the cockpit), Better Pushback (gate departures) — plus everything added in Waves 1-3.

The wave's centerpiece aircraft is the **Boeing 737-800** (stock X-Plane 12 aircraft — no card needed; confirm it is present in your install's aircraft menu).

Per project rule, wave-specific mods come from you — cards below are **pending addition**, with the fields this guide's rules demand. Anything you do not supply stays “**user to provide**”.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Wave 4: Mechanics

The 737-800 is the first aircraft in this guide where the machine is genuinely easier to fly than the operation around it. It is a well-behaved, highly automated airliner that will fly a perfect ILS by itself; your job is the airline operation — the briefing, the fuel, the automation, and the judgment — that surrounds every leg. This chapter is about the systems and the thinking that make that operation work.

Jet Aerodynamics & Performance

A jet is an energy aircraft. The 737 flies high and fast because it carries enormous energy — altitude and speed — and the whole art of flying it is managing that energy smoothly. The takeoff sets the terms: the FMC computes V_1 , V_r , and V_2 from the weight, the runway, and the weather — the go/no-go speed, the rotation speed, and the initial climb speed — and they are a contract. Before V_1 , the takeoff can stop; past V_1 , it continues, because there is no runway left to stop on. Brief the speeds, set the thrust, and honor the contract.

The descent is where the energy game is won or lost. A jet trades speed for altitude and altitude for speed, and arriving high and fast is the classic jet mistake — the one that turns a comfortable arrival into a slammed-on speedbrakes, high-energy approach. The FMC computes a top of descent and a vertical path; fly it, and the approach is set up before the airport appears. The high-altitude cruise adds a second layer: at 35,000 feet the envelope is narrow, with the margins between stall and overspeed compressed, and the autopilot is your friend — the skill is in knowing the envelope through the FMC's speeds and never letting the aircraft drift toward its edges.

The FMC

The Flight Management Computer is the cockpit brain: route, performance, and vertical profile in one box, and the box is only as good as its data. The discipline of the airline pilot is to program it completely — the plan from the briefing, the weights and fuel, the departure and arrival, the approach — then cross-check what you entered against the plan you briefed, and then fly what the box flies while never stopping to verify it. LNAV and VNAV are the modes that matter: lateral navigation flies the route turn by turn, and vertical navigation flies the profile, leveling at the constraints on the route and starting the descent at the computed point. The box is a proposal until it is a fact — the mode annunciation, the needles, and the map are the evidence, and the pilot who watches the evidence is the pilot the box cannot surprise.

The Airline Operation

The airline flight is won before the pushback, in the briefing: the weather at both ends, the fuel, the alternates, and the plan for the whole day, done calmly on the ground rather than discovered in the air. The aircraft then executes — the FMC flies the SID, the route, the STAR — and the crew manages: flows and checklists, callouts and readbacks, and the sterile cockpit, the rule that below ten thousand feet the talk is flying. Even in single-pilot practice, say the callouts out loud; the habit is the safety. The approach is the summit: brief the plate, arm the approach, monitor the capture, and fly the manual flare at the end — the transition that separates operators from passengers.

Fuel, Weight & Contingency

Fuel on an airliner is a legal document, not a guess. Trip fuel for the route, reserves, alternate fuel, and contingency are computed and defended: the fuel page is the cockpit's most important display on a busy day, and every deviation from the plan has a price that the fuel page shows. The alternate is the plan's spine — every leg has one, briefed and loaded, because destinations go below minimums and ATC (or the sim) vectors you around. When it happens, the fuel page and the decision clock are your instruments: the hold, the diversion, the margin — all of it was planned on the ground, and the airborne decision is just the execution.

Wave 4: Content

Content for the airline wave: the network — hub airports, the routes the 737 flies, and scenery that makes a full airline operation feel real. Anything that adds an airport, a route, or a reason to run a schedule belongs here.

Per project rule, content mods come from you — cards below are **pending addition**.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Wave 4: Graphics

The airline wave's visual needs are practical: weather that makes the operation mean something, night lighting for late arrivals, and readability at the high workload moments. The foundational toolset (AutoOrtho, X-Camera) stays in play.

Per project rule, visual mods come from you — cards below are **pending addition**.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Wave 5: The Heavy Ace

Wave 5: Strategy

The Story So Far

The Heavy Ace. The line years ended with a transfer you did not expect: the cargo division. The company's Boeing 777 Freighter flies the world's night — pallets of medicine, cars, and tomorrow's economy, on long-haul tracks across oceans and empty sky. It is the largest, most automated, and most demanding aircraft you will ever fly: a hundred tonnes of payload, a flight deck of screens, and legs measured in hours and continents. The captain's seat on the 777F is the top of the ladder in Ace's Realm — everything you learned in six waves, distilled into one very large, very patient airplane that asks you to think ahead in hours, not minutes. The ace's final exam is not flying it. It is **running** it.

Strategy for the Wave

Wave 5 is **heavy long-haul operations**: the aircraft flies itself; the operation — payload, fuel, oceans, and the night — is the flight.

1. **Think in hours** — long-haul is strategy at continental scale: the fuel plan, the oceanic route, the alternates, and the crew day are decided before pushback and defended all night. Every decision is made twice: once on the ground, once at 35,000 ft.
2. **The automated flight deck** — the 777's fly-by-wire and advanced FMC fly the airplane superbly; your job is the mode discipline of Wave 4, scaled up: program, cross-check, monitor, and always know what the automation thinks it is doing.
3. **The night** — most long-haul legs are darkness: night flying is a systems game (lights, automation, fuel, and the instruments), and the horizon-less world makes the scan and the plan everything.
4. **Payload is destiny** — the 777F's cargo pays for the flight: weight and balance, payload-fuel trade-offs, and the performance pages are the real cockpit conversation.
5. **Aircraft addon** — this wave's aircraft is the **777F** you supplied as the wave's study aircraft — a payware freighter addon (developer/URL: user to provide; see the Modlist section for the card and its status).

Progression Goals

1. Complete a full long-haul leg: payload/fuel briefing, FMC program, oceanic routing, night cruise, and a CAT II/III-appropriate arrival.
2. Manage the fuel day: plan trip + reserve + alternate, and fly a diversion from the plan.
3. Run an oceanic crossing with the long-haul routing tools and position reporting the track demands.
4. Fly a low-visibility arrival using the aircraft's automation to minimums.
5. Graduate: the ace's certificate is one perfect long-haul night — cargo on time, fuel to spare, and the calm of someone who planned for everything.

Mission Ideas

1. **The red-eye freight** — a night crossing with a payload deadline: the classic 777F mission.
2. **The ocean diversion** — an engine or systems scenario mid-ocean: the diversion plan was made on the ground; now it gets flown.
3. **The minimums arrival** — a foggy destination at the edge of the aircraft's approach capability: brief, fly, and decide like a captain.

How to Fly: The Long Haul

The 777F day, step by step. A long-haul freighter flight is won on the ground and flown at night: the briefing, the program, the ocean, and the arrival. The aircraft's manual has the exact numbers; this

chapter teaches the rhythm, and the rhythm of a fifteen-hour day is what makes the heaviest airplane in the guide feel like the most restful one to fly.

The Briefing Ritual

The long-haul briefing is where the flight is decided, and it is a ritual because everything after it is execution. Start with the payload: the cargo, its weight, its position, and the center of gravity it produces — the 777F flies a spreadsheet, and the spreadsheet decides the performance. Then the fuel: trip fuel, reserves, alternate fuel, and contingency, built as a legal document rather than a guess. Then the weather at both ends, and the minimums at the far one — the place where the day can go sideways is known before departure. Then the plan: the route, the oceanic track if there is one, and the diversion plan. The whole night is decided here, calmly, on the ground, so that in the air there is nothing left to decide that has not already been decided well.

The FMC Program

With the briefing done, the boxes get the plan. Load the route — the same plan LittleNavMap built during the briefing — through the waypoints, the oceanic entry and exit, the arrival, and the approach. Enter the performance: the weights and fuel into the performance pages, and the box computes the V-speeds and the thrust settings for the whole day. Then cross-check, twice: the LEGS page against the briefing, leg by leg, because a long-haul mistake is a **long** mistake, and the box gets no free pass. Set the automation for the day — the auto-throttle armed, the targets set, the approach armed for the far end — and the aircraft is ready for a flight that will outlast your entire previous career in hours.

The Oceanic Track

The ocean is crossed on a track: the organized route over the water, with its entry fix, its exit fix, and its reporting points, loaded and checked before entry. Report position on the schedule the track demands — over an ocean, the position report is the radar, and the track only works if everyone on it plays by the clock. At the halfway and at the exit, check the fuel: burn against plan, and the diversion options re-verified, because the fuel page is the cockpit's most important screen at night over water. And carry the drift-down plan in your head and in the fuel: if an engine stops mid-ocean, the aircraft settles to a single-engine altitude, and the plan for that was made on the ground and loaded into the fuel. It is planning, not pessimism — it is what lets you enjoy the quiet.

The Night Cruise

The quiet hours are the real work. The autopilot flies, the fuel burns, and the crew manages the systems and the plan: hourly checks of the fuel page, the position, and the weather picture, with the rhythm of the crew day — rest in rotation, sharpness saved for the arrival. The sim may not need rest, but the habit of managing the day does, and the habit is what the airline version of you would thank you for. As the hours pass, the top of descent arrives, computed and confirmed: the arrival begins an hour before the airport appears, with energy, weather, and the approach all starting to come together. The long night ends the way it began — with the plan.

The Minimums Arrival

The far end of the day is often weather, and the arrival is the same briefing as every wave, at the end of a very long day: the setup, the path, the minimums, the missed — briefed from the plate in AviTab while the aircraft descends. In low visibility, the automation flies the approach to very low minimums and the crew monitors like hawks: needles, speed, mode annunciations, because the automation is powerful and the monitoring is the job. At the decision altitude, the call: the runway environment in sight, or the missed — the missed is the procedure you briefed twelve hours ago, and flying it is not a disappointment, it is the plan. And when the wheels are down and the reverse is in, the dock: the

cargo doors, the payload off, and the night's promise kept. The freighter's day ends with the cargo where it was promised, which is the only scoreboard that matters.

The Debrief

Every long-haul flight deserves a debrief, and the long-haul debrief is a numbers conversation: fuel burn against plan, time against plan, every decimal a lesson about the wind, the weight, or the planning. Then the decisions: the diversions considered, the minimums respected — the log of judgment, not just of flight time. And the logbook itself: six waves of entries, from the first circuit at Cedar Harbor to the transatlantic night. The ace's logbook is the point of Ace's Realm — every entry a flight that was planned, flown, and reviewed, and every review a little more of the calm that the heavy iron demands.

Wave 5: Modlist

Wave 5's centerpiece is a **user-supplied aircraft addon**: the Boeing 777 Freighter. It is the final and most demanding wave, and the capstone of the foundational toolset plus every system added in Waves 1-4.

FlightFactor 777-200ERF v2 Ultimate (777F)

Dependencies: user to provide

Impact: Adds the 777-200ERF widebody freighter as this wave's study aircraft — the largest and most complex aircraft in the guide.

Notes: User-supplied aircraft (payware addon). Current version and any dependencies: user to provide. Per project rule, nothing here validates compatibility — confirm against 12.4.3 with the release-date heuristic.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Wave 5: Mechanics

The 777F is the largest, most automated, and most demanding aircraft in this guide — and the least demanding to fly. The machines of this wave fly themselves superbly; the operation around them — payload, fuel, oceans, and the night — is the flight. This chapter is about thinking in hours and continents, which is the real curriculum of the heavy wave.

The Heavy Aircraft

The 777's flight controls are fly-by-wire: your inputs are commands to a computer that moves the control surfaces and protects the envelope — it will not let you stall or overspeed the aircraft in normal flight. The protection manages limits, not judgment: the aircraft keeps itself inside the envelope, and you keep the operation inside the plan. The second system to internalize is weight and balance. A hundred tonnes of payload is a spreadsheet with wings: the zero fuel weight, the fuel load, and the center of gravity decide how the aircraft actually flies, and the performance pages are where that conversation happens. The briefing is where the flight is won, because the spreadsheet was correct before the engines started.

The Long-Haul Flight Deck

The 777's flight deck is the FMC at continental scale: route, performance, and vertical profile for a leg measured in hours, programmed from the briefing and flown with the mode discipline of Wave 4, scaled up. The modes themselves are the same ideas — managed versus selected, LNAV and VNAV, the auto-throttle — but the stakes are the size of the leg: a programming error found at the oceanic entry is a different problem than one found at the destination. The constant question — **what is the automation doing, and why?** — is the whole job, and the fuel page is the cockpit's primary display: burn, reserves, and the diversion fuel are the numbers that decide every option. Defend the plan; know the price of every deviation.

Oceanic & Long-Range Operations

Over an ocean there is no radar, and the structure that replaces it is the organized track: routes across empty water with mandatory position reporting, flown exactly as published, because the report is the radar. ETOPS thinking underlies it — extended-range operations assume an engine can stop mid-ocean, so the diversion airport, the fuel to reach it, and the single-engine drift-down plan are all decided before departure. It is planning, not pessimism: long legs have few alternates, and the destination alternate, the en-route alternates, and the fuel to make any of them are the briefing's spine. The ocean is flown the way it is planned — on the track, on the clock, and with the fuel in hand.

Night & Low-Visibility Flying

Most long-haul legs are darkness, and night flying is a systems game: lights, instruments, and automation, flown on the scan and the plan with the horizon gone. The low-visibility arrival is the summit — CAT II and III approaches that let the autopilot fly to very low minimums, with your job being the briefing, the monitoring, and the decision at the decision altitude. The hand-fly below is optional; the call is not. And the crew day is its own system: long-haul is crew resource management, rest in rhythm, and the sterile cockpit during the busy phases — the ace flies the day, not just the leg, and the day is fifteen hours long.

Emergency Discipline

The abnormal procedures of the heavy aircraft are flows and checklists, flown calmly in order: engine, systems, pressurization — the sim gives you the scenarios, the procedure gives you the answer. The diversion decision at 35,000 feet over an ocean is the professional's reflex, made with the fuel plan already in hand, not improvised at the moment. And the calm itself is the point of the whole wave: everything in this chapter exists to make the abnormal ordinary, so that when something goes wrong, the boredom of a well-practiced procedure takes over — because the planning already happened, on the ground, hours ago.

Wave 5: Content

Content for the heavy wave: the long-haul network — hubs, cargo terminals, and the world the 777F crosses at night. Anything that makes a continent-scale operation feel real belongs here.

Per project rule, content mods come from you — cards below are **pending addition**.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Wave 5: Graphics

The heavy wave's visual needs: night lighting (most long-haul is dark), weather that gives the operation teeth, and low-visibility presentation for the minimums arrivals. The foundational toolset stays in play.

Per project rule, visual mods come from you — cards below are **pending addition**.

user to provide

Dependencies: user to provide

Impact: user to provide

Notes: user to provide

Glossary

Flight Basics

1. **VFR / IFR** — Visual Flight Rules (fly by looking outside, good weather) vs Instrument Flight Rules (fly by instruments, any weather). Waves 0-1 are VFR-first; Waves 2+ add IFR.
2. **Traffic pattern** — the standard rectangular circuit around an airport: upwind, crosswind, downwind, base, final.
3. **Downwind / Base / Final** — the three legs before landing: parallel to the runway (downwind), turning across the threshold (base), and the final straight-in approach.
4. **Rotate** — pulling back on the yoke to lift off at takeoff speed (V_r).
5. **Flare** — the gentle nose-up just before touchdown that lets the wheels settle, not slam.
6. **V-speeds** — design speeds: V_s (stall), V_r (rotation), V_x/V_y (best angle/best rate climb), V_1/V_2 (takeoff decision/initial climb in jets). The numbers live in each aircraft's documentation.
7. **IAS / TAS** — Indicated vs True Airspeed: what the instrument shows vs actual speed through the air (they differ with altitude and temperature).
8. **AGL / MSL** — height Above Ground Level vs Above Mean Sea Level (altimeter setting reference).
9. **QNH** — the altimeter pressure setting that makes the altimeter read field elevation on the ground; set it from ATIS or the local METAR.
10. **ATIS** — recorded airport information (weather, runways in use) played on a loop.
11. **METAR / TAF** — the routine weather report / the terminal forecast.
12. **Squawk / Transponder** — the aircraft's identification code and the transponder that answers ATC radar; 7700 = emergency, 7500 = hijack, 7600 = radio failure.
13. **Cold & dark** — an aircraft parked with everything off; starting from cold & dark means doing the full start flow yourself.

Instruments & Avionics

1. **PFD / MFD** — Primary Flight Display (attitude, airspeed, altitude, heading) and Multi-Function Display (map, flight plan, systems) on glass-cockpit aircraft.
2. **ASI / Altimeter / VSI / Attitude / Heading** — the “six pack”: airspeed, altitude, vertical speed, artificial horizon (attitude), heading indicator, plus turn coordinator.
3. **FMS / FMC** — Flight Management System / Computer: the route computer that flies waypoints, procedures, and vertical profiles.
4. **Autopilot modes** — HDG (heading), ALT/ALT hold, V/S (vertical speed), LNAV/VNAV (lateral/vertical navigation along the route), APP (approach tracking), and in jets FLCH and managed/selected modes.
5. **ILS** — Instrument Landing System: localizer (lateral) + glideslope (vertical) guidance for approaches; CAT I/II/III are categories of decreasing minimums.
6. **VOR / DME / NDB** — classic navaids: VHF omnidirectional range (radial navigation), distance measuring equipment, and non-directional beacon (A/N needle).
7. **RNAV / GPS** — area navigation: flying direct between waypoints by satellite rather than toward a ground navaid.

Airspace & Procedures

1. **SID / STAR** — Standard Instrument Departure / Standard Terminal Arrival Route: the published highways out of and into busy airports.

2. **Approach plate** — the chart for an instrument approach: courses, altitudes, minimums, missed-approach procedure.
3. **Holding pattern** — the racetrack you fly while waiting for clearance.
4. **CTAF / UNICOM** — the radio frequency where non-towered airports self-coordinate (“traffic, Cessna one-seven-two, downwind, runway two-four, Bayfield”).
5. **Tower / Ground / Clearance** — the ATC positions at towered airports: clearance delivery, ground (taxi), tower (runway), approach/departure, center.
6. **Vectors** — headings ATC assigns you, usually toward a fix or final approach.

Aircraft & Systems

1. **Yoke / Stick** — the control column (pitch and roll); rudder pedals steer on the ground and coordinate turns in the air; trim removes steady control pressure.
2. **Flaps / Slats** — high-lift devices that allow slower, steeper approaches and short landings.
3. **Mixture** — the fuel/air ratio control (piston aircraft): lean as you climb, rich for takeoff.
4. **Torque / ITT** — turboprop instruments: engine torque (power) and turbine inlet temperature (temperature limit — the one to watch on hot days).
5. **N1 / N2** — turbofan compressor speeds: N1 (fan) is the thrust gauge, N2 (core) the “engine running” gauge; thrust is managed against N1 targets.
6. **EGT** — exhaust gas temperature: the turbofan’s critical limit gauge on hot days and at high power.
7. **Beta range / Reverse** — turboprop propeller pitch below flight idle: beta for precise ground control, reverse for stopping.
8. **Pressurization** — the cabin pressure system that lets turbine aircraft fly high without masks; set the cruise altitude and the cabin climbs/descends gently.
9. **Fly-by-wire** — computer-mediated flight controls (modern airliners); the computer protects the envelope.

Weather

1. **Wind / Crosswind** — wind direction/speed (e.g., “240 at 12”) and the component across the runway that you must correct with crab or sideslip.
2. **Ceiling / Visibility** — cloud base and horizontal visibility; the two numbers in most weather minimums.
3. **Icing** — frozen moisture on wings/surfaces; anti-ice and de-ice systems exist for a reason — avoid it early, escape it if you meet it.
4. **Fronts & pressure** — low pressure + fronts = weather; high pressure = fair skies. The sim’s weather engine honors real-world (live) weather if you enable it.

X-Plane & Modding

1. **Scenery pack** — any folder in Custom Scenery/; an airport, an overlay, ortho tiles, or a library. Load order is set by scenery_packs.ini.
2. **scenery_packs.ini** — the text file in Custom Scenery/ listing every pack top-to-bottom; higher = drawn on top.
3. **Ortho** — photoreal ground textures from satellite/aerial imagery (AutoOrtho streams them).
4. **Mesh** — the terrain height field the textures are draped over.
5. **Plugin** — a program that runs inside X-Plane (folder in Resources/plugins/); the foundational tools are plugins.
6. **Livery** — an aircraft’s paint scheme (folder inside the aircraft’s liveries/).
7. **Navdata / AIRAC** — navigation databases; AIRAC is the 28-day update cycle (Navigraph).
8. **Log.txt** — the sim’s log file at the root; the first place to look when something does not load.

9. **FMS plan (.fms)** — the sim's flight-plan file format (LittleNavMap exports it).